Design and Implementation of a Non-Focal Rotman Lens for 360° Beam Steering at 24 GHz

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***Abstract*—This paper presents the design, numerical synthesis, and PCB simulation of a non-focal Rotman lens operating at 24 GHz for 360° beam steering applications. The proposed design utilizes more than four pseudo-focal points to improve beam resolution and reduce phase error across a wider scan range. A MATLAB-based CAD synthesis method was developed to generate beam port locations, dummy ports, and full PCB geometry. An antenna array utilizing U-slot patch antenna elements was developed in Ansys HFSS to explore the impacts of a larger bandwidth antenna element on the output beam. The lens and antenna array were implemented on a Rogers 4350b substrate using a microstrip transmission line structure with impedance-matched tapers. The resulting design enables passive beam steering across multiple fixed angles without active phase shifters. This work demonstrates a practical method for implementing near full-scan, non-focal Rotman lenses using generalized beam contour synthesis.**

***Index Terms*—Rotman lens, numerical optimization, non-focal lens, phase shifting, passive beam steering**

I. Introduction

Rotman lenses provide a passive beamforming solution capable of generating multiple simultaneous beams without requiring active phase shifters [1]. By introducing controlled path length differences between beam ports and array ports, the Rotman lens produces the phase progression necessary for beam steering. This approach significantly reduces system complexity, power consumption, and cost compared to active beamforming architectures, making it well-suited for millimeter-wave and 5G applications [2]–[4].

Traditional Rotman lens designs employ circular or elliptical beam contours and three focal points to minimize phase error at discrete steering angles. However, this constraint limits the performance between focal points, resulting in phase aberrations and non-uniform beam resolution throughout the scan range [5]. For applications requiring wide-angle or full azimuthal coverage, alternative beam contour formulations can improve performance.

Moreover, conventional designs of Rotman lenses have been restricted to linear or planar arrays with scan angles limited to $\pm 60°$. To address this limitation, Nguyen et al. [6] introduced a 360° scanning circularly symmetric polygon lens, leveraging the symmetry created by a circularly-shaped Rotman lens. The beam and array ports were arranged in a polygonal structure, effectively extending the output beam coverage. The arrangement of beam and array ports in a polygonal shape is not implemented in this design, but instead focuses on a mechanical approach by simulating a stacked polygonal structure. The interleaved beam and array ports are not considered for this design due to the limited angular coverage per lens that will eventually superpose to achieve a 360° scan.

In this work, a non-focal Rotman lens operating at 24 GHz is designed using a generalized elliptical beam contour. An operating frequency of 24 GHz was chosen to optimize the Rotman lens for the highband 5G, the most common generation of global wireless cellular technology in modern day usage. Unlike classical tri-focal designs, the proposed approach distributes many pseudo-focal beam ports along an elliptical contour to improve phase uniformity and increase angular resolution. In MATLAB, a numerical synthesis method was implemented to compute beam port locations, dummy ports, and lens geometry. The resulting structure was implemented using microstrip transmission lines and a linear antenna array on a Rogers 4350b substrate and prepared for PCB fabrication [3], [7].

This work focuses on the mathematical formulation, numerical synthesis, and physical implementation of the lens geometry, providing a complete design pipeline from electromagnetic theory to PCB realization.

II. Methodology

A. Rotman Lens Geometrical Model

The Rotman lens was designed based on equal electrical path length to ensure true time-delay phase shift and beamforming. The structure of the Rotman lens consists of a beam port contour, an array port contour, and a set of transmission lines connecting the lens to the antenna elements.

The lens geometry is designed to satisfy the condition:

$$L_{b,ij} + W_j - N_{y,j} \sin\theta_i = L_{0,i}$$

where $L_{b,i,j}$ is the distance between the ith beam port and the jth array port, W_j is the length of the transmission line connected to the jth array port, $N_{y,j}$ is the position of the jth array element along the y-axis, θ_i is the scan angle of the ith beam port, and $L_{0,i}$ is the reference focal path length of the ith array port.

B. Design Parameters

The operating frequency was set to 24 GHz. The spacing between each array element was chosen to be $0.5\lambda_0$, where λ_0

denotes the free-space wavelength at the operating frequency. The lens configuration consists of eleven beam ports and eleven array ports to achieve sufficient angular resolution across the scan angle range. The maximum scan angle was set to be $\pm 30°$. Additionally, to mitigate reflections and enhance impedance matching within the lens, dummy ports were incorporated along the boundaries connecting the beam and array contours.

C. Numerical Optimization

The lens contour coordinates were obtained by numerically solving the nonlinear path length equation in MATLAB. The optimization minimized the objective function of the total path length error across all array and beam ports:

$$\text{minimize} \quad J = \sum_{n} \left(\phi_n - \phi_n^{ideal}\right)^2$$

where ϕ_n is the calculated phase of the n_{th} beam port and ϕ_n^{ideal} is the ideal phase progression for a given scan angle.

The constrained nonlinear optimization was performed using the `fmincon` function in MATLAB, which employs a gradient-based interior point algorithm to minimize a nonlinear objective function subject to linear and nonlinear constraints and variable bounds. The algorithm solves the Karush-Kuhn-Tucker (KKT) optimality conditions iteratively while maintaining the parameter values and candidate solutions within the feasible region until convergence. The optimization constraints were determined with respect to physical feasibility of the contour coordinates, positive transmission line lengths, and appropriate port spacings. The optimization tolerance was set to 10^{-9}.

D. Taper and Feedline Design

The lens required exponential tapers and microstrip feedlines to ensure optimal signal transfer between the beam ports, array ports, and antenna array. A series of tapers, adjustable based on the starting and terminating impedances, are placed at every lens port, including the dummy ports. These tapers transition between a very low arbitrary impedance and terminate in 50 ohms, which transitions from a high to low width. This ensures a proper transition into the microstrip feedline, especially at the top and bottom contours of the lens, absorbing input power from the beam ports while minimizing reflections, which maximizes return loss. A series of transmission lines have also been placed between the array ports and the corresponding antennae of the antenna array. Meandering geometries were utilized when necessary to account for spatial constraints while maintaining the desired electrical lengths.

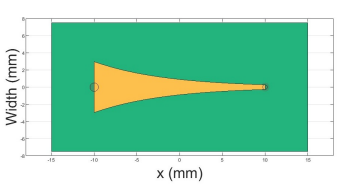


Fig. 1. Exponential taper geometry showing gradual transition from 1 Ohm to 50 Ohms

E. Antenna Array Design

The antenna array is responsible for the gain, directivity, and efficiency of the output radiation beam. The array is composed of 11 linear U-slot patch antenna elements designed to resonate at 24 GHz. A sample U-slot antenna was designed and simulated in Ansys HFSS (High-Frequency Structure Simulator) to optimize the antenna geometry. A U-slot patch element was selected to observe the effect of a widened bandwidth on the 24 GHz frequency range and its compatibility with identical elements in an array setting.

The width of the patch antenna was dictated by

$$W = \frac{c}{2f_0}\sqrt{\frac{2}{\epsilon_r + 1}}$$

where f_0 is the target frequency of 24 GHz and ϵ_r is the relative permittivity of the substrate (3.48 for RO4350b). The ideal length of the patch, which primarily determines the resonant frequency, was set to half a guided wavelength, found using the effective permittivity of the substrate and a 24 GHz wavelength in free space. Taking into account fringing field extensions, the final calculations resulted in a patch width of 4.172 mm and a length of 3.22 mm.

The vertical arms of the U-slot region widen the patch's radiation bandwidth through constructive interference with the main patch currents. Using a relationship of $\frac{\lambda_g}{4}$, the vertical arms were found to be 1.62 mm.

A microstrip feed line inset into the patch received a signal on a perpendicular plane and carried to the patch. A parametric sweep was used to determine the position of the feed line and the inset dimensions, with a final length of 2.3 mm and a width of 0.3 mm.

The array was created using the lattice pairs boundary option in HFSS to allow for uniform shapes and radiation boundaries amongst the elements. Each element is 6.25 mm apart ($\frac{\lambda_g}{2}$) from the center of one element to the center of the adjacent one.

III. Results

A. Lens Geometry/Optimization Data

The root mean square (RMS) phase error of the Rotman lens was evaluated as a function of external scan angle θ for multiple lenses, each with different number of perfect/pseudo-focal points. Figure 8 compares lens designs with 3 perfect focal points to 11 pseudo focal points over a $\pm 30°$ scan angle range.

Across all lens designs, the phase error varies with beam port angle and remains symmetrical about $\theta = 0°$, consistent with the symmetric geometry of the lens.

In the conventionally designed lens with 3-foci, the RMS is minimized at $\theta = 0°$, and increases rapidly with scan angle and reaches its peak before approaching zero at $\theta = \pm 30°$. This indicates that the traditional design is strongly optimized for operating at only the three perfect focal points, but exhibits substantial phase degradation at other points along the beam contour.

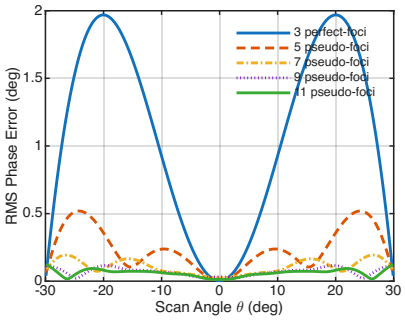


Fig. 2. RMS phase error as a function of external scan angle θ for Rotman lenses synthesized with 3, 5, 7, 9, and 11 pseudo-foci.

An important trend is that optimized lenses with increasing number of pseudo-foci redistributes the RMS error more evenly across the scan angle range. Especially, for 9 and 11 pseudo-foci, the RMS curve becomes more uniform in the central region of the scan range ($\pm 10°$ to $\pm 20°$), and the error peaks are reduced significantly, indicating improved edge-angle performance. However, diminishing return effect is present as the performance gained per pseudo-foci added reduces and the aggregate RMS phase error approaches an asymptotic value as shown in Fig. 3. The aggregate RMS error mentioned in the figure is defined as the root-mean-square of the RMS phase error profile across all scan angles, and serves as a singular scalar value for comparing lens designs with different number of pseudo-focal points.

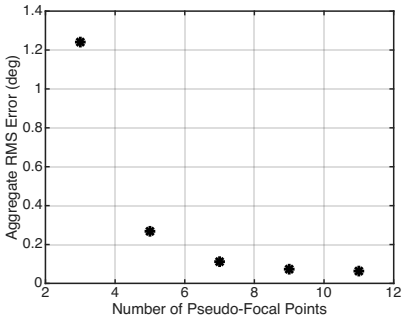


Fig. 3. Aggregate RMS phase error as a function of the number of pseudo-focal points for a Rotman lens operating at 24 GHz.

B. S-parameters

Shown below are the S_{11} parameter values for every beam ports at 24 GHz. The outer ports show a higher return loss and less power reflection than the inner ports.

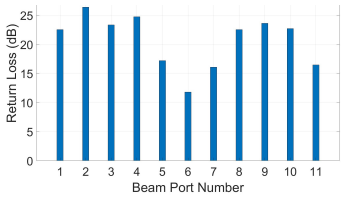


Fig. 4. Return loss after every beam port (1 to 11) is excited

C. Antenna Radiation Pattern

A simulation of an individual U-Slot patch antenna element was run in HFSS using a vacuum boundary with a target resonant frequency of 24 GHz.

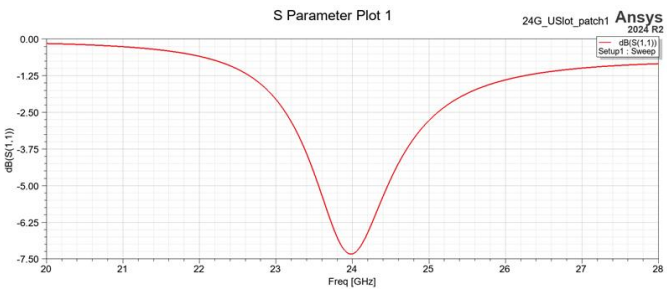


Fig. 5. S11 Parameter Plot of an individual U-slot patch antenna element

The S_{11} parameter plot of an individual antenna element displays a null at 24 GHz, with a 3 dB bandwidth extending from approximately 23.34 GHz to 24.68 GHz. This is representative of the larger bandwidth that the U-slot provides. The minimum of -7.33 dB indicates that 82.2% of the incident power is absorbed and 17.8% is reflected back, suggesting improvements in efficiency and performance. Ideally, a return loss minimum of -10 dB or greater in absolute value is aimed for so that less than 10% of incident power is reflected. A closer investigation into the impedance matching of the microstrip feed line to the antenna is required to resolve the low return loss.

D. Linear Array Verification

A simulation of the antenna array was carried out assuming an idealized linear phase distribution and uniform magnitude weights for each of the elements, verifying solely the array fed with ideal signals. The HFSS Finite Array Beam Angle Calculator was applied to predict the main beam angle upon simulation. As shown in the figure, the difference between the main lobe and the highest side lobe is approximately 9dB.

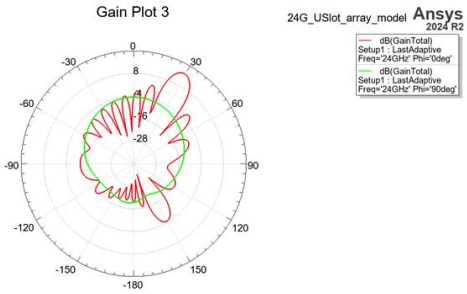


Fig. 6. Total Gain (dB) plot of the antenna array across the azimuth (ϕ) plane

The green curve represents $\phi = 90°$, displaying radiation behavior horizontally around the array and $90°$ away from the main lobe. This plot indicates a broad beam width in that plane due to the circular shape, meaning that the antenna array radiates uniformly in all directions on the horizontal

plane. Therefore, it is expected that the antenna array will provide nearly equal omnidirectional coverage. The red curve represents $\phi = 0°$, the array factor pattern. The peak gain of the main lobe is +10 dB in the +30° direction. There are some strong side lobes, especially in the +150° direction, which indicates poor radiation efficiency. This suggests that some of the elements may be out of phase with one another, requiring a deeper inspection into the impedance matching of the feed line to each element and whether the substrate could be contributing dielectric losses.

The simulation of the antenna array was carried out assuming idealized linear phase tapers and uniform magnitude weights for each of the elements.

E. Full System Integration and Testing

Shown below are the far-field radiation pattern for every beam port and the 3D view of the final lens design. At high steering angles, there was significant degradation, caused by improper feed line scaling, which will be addressed in future work.

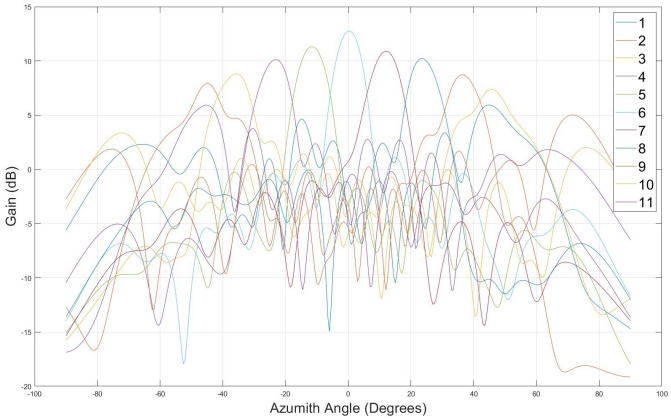


Fig. 7. Far field radiation pattern (ports 1 to 11)



Fig. 8. 3D model of final lens

IV. Conclusion

This work presented the design and numerical synthesis of a non-focal Rotman lens using an elliptical beam contour for operation at 24 GHz.

A numerical-based synthesis method was developed in MATLAB to compute beam and array port positions, dummy ports, and PCB geometry. The resulting design enables passive beam steering without active phase shifters.

The proposed lens design, with 11 pseudo-focal points, demonstrates reduced phase error and improved performance, flexibility, and scan resolution compared to conventional Rotman lens designs.

The antenna array design with a U-slot patch antenna basis provides greater insight on the use of the U-slot design in array settings. The increased bandwidth needs to be verified through further integration simulations and testing.

Future work includes fabricating the final PCB and integrating the physical antenna array at the array ports to validate the expected S-parameters and radiation patterns from the HFSS simulations. Long-term applications include preventing dead zones that result from higher frequencies within 5G/6G, while retaining higher data rates, reduced latency, and increased bandwidth.

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